

# AI Agent Design

ServiceGuard AI

## 1. Agent purpose

The agent investigates operational questions by selecting bounded tools, aggregating evidence, and producing a structured recommendation. It is not an unrestricted autonomous operator.

## 2. Tools

| Tool                    | Purpose                               |
|-------------------------|---------------------------------------|
| query_service_telemetry | Read current/recent metrics           |
| get_dependency_health   | Inspect dependency/context signals    |
| search_runbook          | Retrieve approved procedural evidence |
| predict_risk            | Call versioned ML model               |
| get_incident_history    | Inspect similar historical incidents  |

## 3. State

```
messages
request_id
user_id
service_id
incident_id
facts
tool_results
risk_assessment
runbook_evidence
recommended_actions
approval_required
```

## 4. Agent graph

```
START
↓
Reasoner
↓
Tool selection
↓
Tool execution
↓
Evidence aggregation
↓
Risk / SOP checks
↓
Reasoner
↓
Structured answer
↓
END
```

## 5. Guardrails

- Tool allowlist.
- Read-only operational tools by default.
- Structured JSON output.
- Maximum tool-call count.
- Timeouts on every external operation.

- No arbitrary SQL generation/execution.
- No credential access.
- Human approval for high-impact actions.
- Explicit uncertainty when evidence is incomplete.

## 6. Example answer contract

```
{
  "summary": "...",
  "riskLevel": "HIGH",
  "riskScore": 0.91,
  "facts": [...],
  "evidence": [
    {"type": "telemetry", "id": "..."},
    {"type": "runbook", "id": "..."}
  ],
  "recommendedActions": [...],
  "requiresHumanApproval": true
}
```